

ECON 0100 | Homework B5 | Market Changes

Due: Sunday, Oct. 4

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \left(\frac{1}{2}\right)Q_d$$

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \left(\frac{1}{2}\right)Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Price Elasticity

- a) Use the midpoint method to find the elasticity of demand when the price changes from 10 to 30. _____
- b) Use the midpoint method to find the elasticity of demand when the price changes from 40 to 60. _____
- c) Use the midpoint method to find the elasticity of demand when the price changes from 70 to 90. _____
- d) Is demand elastic, unit elastic, or inelastic in a), b), and c)? _____

e) Use the midpoint method to find the elasticity of supply when the price changes from 40 to 60. _____

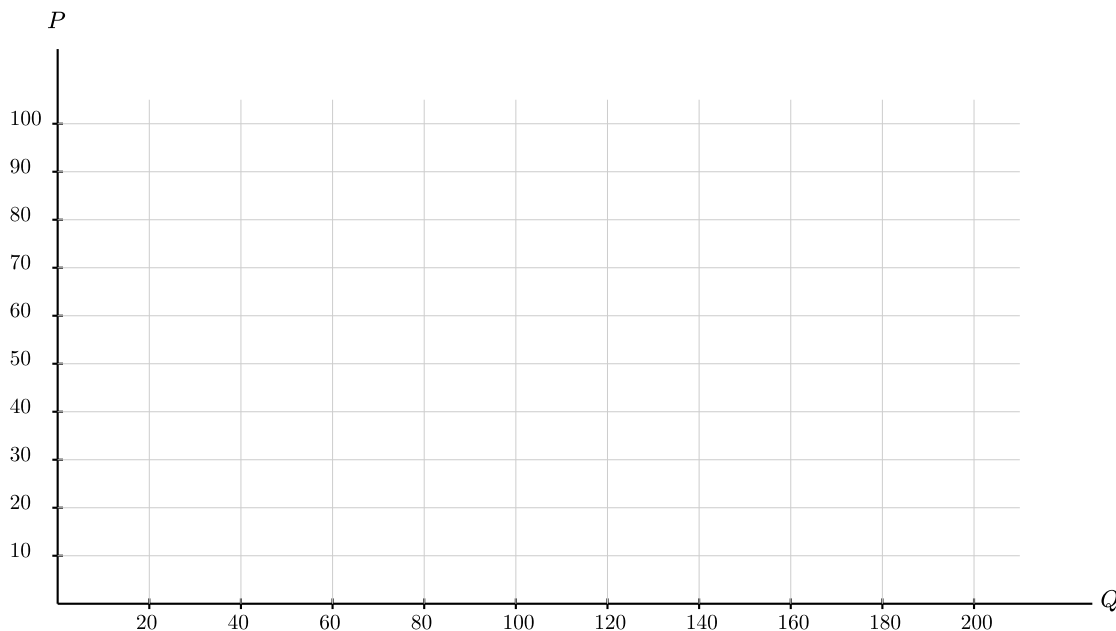
Q2 | Supply & Demand Shifters

Butter beer has culturally been associated with a good time. This all changed when a high profile wizard went viral slamming the taste. At the same time an ingredient of the drink, butter extract, became incredibly difficult for butter beer makers to acquire. Use a graph to discuss the effect these two events had on the butter beer market.

- a) After the viral post, the demand curve: *shifted in, stayed the same, shifted out*
- b) After the butter extract shortage, the supply curve: *shifted in, stayed the same, shifted out*

Separately, suppose wages rise across the wizarding world.

- c) If butter beer is a normal good, the demand curve: *shifted in, stayed the same, shifted out*
- d) If butter beer is an inferior good, the demand curve: *shifted in, stayed the same, shifted out*



Q3 | Comparative Statics

After the butter extract shortage alone, the supply curve became:

$$P = 30 + \left(\frac{1}{2}\right)Q$$

a) Equilibrium price: _____ Equilibrium quantity: _____

Without using numbers, how did the two events together (the viral post and the butter extract shortage) change the market?

b) Prices: *increased, stayed the same, decreased, indeterminate*

c) Quantity: *increased, stayed the same, decreased, indeterminate*

After both events, the demand curve became:

$$P = 80 - \left(\frac{1}{2}\right)Q$$

d) Equilibrium price: _____ Equilibrium quantity: _____

